



General Linear Regression Analysis

The k -Variable Linear Regression Model

If we generalize the two- and three-variable linear regression models, the k -variable population regression function (PRF) model involving the dependent variable Y and $k - 1$ explanatory variables $X_2, X_3, \dots, X_k$ may be written as

$$\text{PRF: } Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i \quad i = 1, 2, 3, \dots, n \text{(1)}$$

where β_1 = the intercept, β_2 to β_k = partial slope coefficients, u = stochastic disturbance term, and $i = i$ th observation, n being the size of the population. The PRF (1) is to be interpreted in the usual manner: It gives the mean or expected value of Y conditional upon the fixed (in repeated sampling) values of $X_2, X_3, \dots, X_k$, that is, $E(Y | X_{2i}, X_{3i}, \dots, X_{ki})$.

The k -Variable

Equation (1) is a shorthand expression:

$$Y_1 = \beta_1 + \beta_2 X_{21} + \dots + \beta_k X_{k1}$$
$$Y_2 = \beta_1 + \beta_2 X_{22} + \dots + \beta_k X_{k2}$$

Equation (1) is a shorthand expression for the following equations:

$$\begin{aligned} Y_1 &= \beta_1 + \beta_2 X_{21} + \beta_3 X_{31} + \cdots + \beta_k X_{k1} + u_1 \\ Y_2 &= \beta_1 + \beta_2 X_{22} + \beta_3 X_{32} + \cdots + \beta_k X_{k2} + u_2 \\ &\vdots \\ Y_n &= \beta_1 + \beta_2 X_{2n} + \beta_3 X_{3n} + \cdots + \beta_k X_{kn} + u_n \end{aligned} \quad (2)$$

Let us write the system of equations (2) in an alternative but more illuminating way as follows:²

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{21} & X_{31} & \cdots & X_{k1} \\ 1 & X_{22} & X_{32} & \cdots & X_{k2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{2n} & X_{3n} & \cdots & X_{kn} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (3)$$

$$\underset{n \times 1}{\mathbf{y}} = \underset{n \times k}{\mathbf{X}} \underset{k \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\mathbf{u}}$$

The k -Variable Linear Regression Model

where $\mathbf{y} = n \times 1$ column vector of observations on the dependent variable Y
 $\mathbf{X} = n \times k$ matrix giving n observations on $k - 1$ variables X_2 to X_k ,
the first column of 1's representing the intercept term (this matrix is also
known as the **data matrix**)
 $\boldsymbol{\beta} = k \times 1$ column vector of the unknown parameters $\beta_1, \beta_2, \dots, \beta_k$
 $\mathbf{u} = n \times 1$ column vector of n disturbances u_i

Using the rules of matrix multiplication and addition, the reader should verify that systems (2) and (3) are equivalent.

System (3) is known as the *matrix representation of the general (k -variable) linear regression model*. It can be written more compactly as

$$\begin{array}{ccccccc} \mathbf{y} & = & \mathbf{X} & \boldsymbol{\beta} & + & \mathbf{u} & \\ n \times 1 & & n \times k & k \times 1 & & n \times 1 & \end{array} \quad (4)$$

Where there is no confusion about the dimensions or orders of the matrix $\mathbf{X}$ and the vectors $\mathbf{y}$, $\boldsymbol{\beta}$, and $\mathbf{u}$, Eq. (C.1.4) may be written simply as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad (5)$$

Assumptions of the Classical Linear Regression Model in Matrix Notation

Scalar Notation

1. $E(u_i) = 0$, for each i (3.2.1)
2. $E(u_i u_j) = 0 \quad i \neq j$ (3.2.5)
 $= \sigma^2 \quad i = j$ (3.2.2)
3. $X_2, X_3, \dots, X_k$ are nonstochastic or fixed
4. There is no exact linear relationship among the X variables, that is, no multicollinearity (7.1.9)
5. For hypothesis testing, (4.2.4)
 $u_i \sim N(0, \sigma^2)$

Matrix Notation

1. $E(\mathbf{u}) = \mathbf{0}$
where $\mathbf{u}$ and $\mathbf{0}$ are $n \times 1$ column vectors, $\mathbf{0}$ being a null vector
2. $E(\mathbf{u}\mathbf{u}') = \sigma^2 \mathbf{I}$
where $\mathbf{I}$ is an $n \times n$ identity matrix
3. The $n \times k$ matrix $\mathbf{X}$ is nonstochastic, that is, it consists of a set of fixed numbers
4. The rank of $\mathbf{X}$ is $p(\mathbf{X}) = k$, where k is the number of columns in $\mathbf{X}$ and k is less than the number of observations, n
5. The $\mathbf{u}$ vector has a multivariate normal distribution, i.e., $\mathbf{u} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$

Assumptions of the Classical Linear Regression Model in Matrix Notation

- Assumption 1 given in Eq. (1) means that the expected value of the disturbance vector $\mathbf{u}$, that is, of each of its elements, is zero. More explicitly, $E(\mathbf{u}) = \mathbf{0}$ means

$$E \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} E(u_1) \\ E(u_2) \\ \vdots \\ E(u_n) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (2)$$

We can write

$$E(\mathbf{u}\mathbf{u}') = E \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} [u_1 \quad u_2 \quad \cdots \quad u_n]$$

Assumptions of the Classical Linear Regression Model in Matrix Notation

where $\mathbf{u}'$ is the transpose of the column vector $\mathbf{u}$, or a row vector. Performing the multiplication, we obtain

$$E(\mathbf{u}\mathbf{u}') = E \begin{bmatrix} u_1^2 & u_1u_2 & \cdots & u_1u_n \\ u_2u_1 & u_2^2 & \cdots & u_2u_n \\ \cdots & \cdots & \cdots & \cdots \\ u_nu_1 & u_nu_2 & \cdots & u_n^2 \end{bmatrix}$$

Applying the expectations operator E to each element of the preceding matrix, we obtain

$$E(\mathbf{u}\mathbf{u}') = \begin{bmatrix} E(u_1^2) & E(u_1u_2) & \cdots & E(u_1u_n) \\ E(u_2u_1) & E(u_2^2) & \cdots & E(u_2u_n) \\ \cdots & \cdots & \cdots & \cdots \\ E(u_nu_1) & E(u_nu_2) & \cdots & E(u_n^2) \end{bmatrix} \quad (2)$$

Assumptions of the Classical Linear Regression Model in Matrix Notation

Because of the assumptions of homoscedasticity and no serial correlation, matrix (2) reduces to

$$\begin{aligned} E(\mathbf{uu}') &= \begin{bmatrix} \sigma^2 & 0 & 0 & \dots & 0 \\ 0 & \sigma^2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \sigma^2 \end{bmatrix} \\ &= \sigma^2 \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} \\ &= \sigma^2 \mathbf{I} \end{aligned} \quad (3)$$

where $\mathbf{I}$ is an $n \times n$ identity matrix.

Assumptions of the Classical Linear Regression Model in Matrix Notation

- Matrix (2) (and its representation given in Eq. (3)) is called the **variance-covariance matrix** of the disturbances u_i
- The elements on the **main diagonal** of this matrix (running from the upper left corner to the lower right corner) give the **variances**
- The elements off the main diagonal give the covariances.
- Note that the variance–covariance matrix is **symmetric**: The elements above and below the main diagonal are reflections of one another.

OLS Estimation

To obtain the OLS estimate of β , let us first write the k -variable sample regression function (SRF):

$$Y_i = \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \cdots + \hat{\beta}_k X_{ki} + \hat{u}_i \quad (1)$$

which can be written more compactly in matrix notation as

$$\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}} \quad (2)$$

and in matrix form as

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{21} & X_{31} & \cdots & X_{k1} \\ 1 & X_{22} & X_{32} & \cdots & X_{k2} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & X_{2n} & X_{3n} & \cdots & X_{kn} \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} + \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \vdots \\ \hat{u}_n \end{bmatrix} \quad (3)$$
$$\begin{matrix} \mathbf{y} & = & \mathbf{X} & & \hat{\boldsymbol{\beta}} & + & \hat{\mathbf{u}} \\ n \times 1 & & n \times k & & k \times 1 & & n \times 1 \end{matrix}$$

where $\hat{\boldsymbol{\beta}}$ is a k -element column vector of the OLS estimators of the regression coefficients and where $\hat{\mathbf{u}}$ is an $n \times 1$ column vector of n residuals.

OLS Estimation

As in the two- and three-variable models, in the k -variable case the OLS estimators are obtained by minimizing

$$\sum \hat{u}_i^2 = \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_{2i} - \cdots - \hat{\beta}_k X_{ki})^2 \quad (4)$$

where $\sum \hat{u}_i^2$ is the residual sum of squares (RSS). In matrix notation, this amounts to minimizing $\hat{\mathbf{u}}' \hat{\mathbf{u}}$ since

$$\hat{\mathbf{u}}' \hat{\mathbf{u}} = [\hat{u}_1 \quad \hat{u}_2 \quad \cdots \quad \hat{u}_n] \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \vdots \\ \hat{u}_n \end{bmatrix} = \hat{u}_1^2 + \hat{u}_2^2 + \cdots + \hat{u}_n^2 = \sum \hat{u}_i^2 \quad (5)$$

Now from Eq. (2) we obtain

$$\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} \quad (6)$$

Therefore,

$$\begin{aligned} \hat{\mathbf{u}}' \hat{\mathbf{u}} &= (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \\ &= \mathbf{y}'\mathbf{y} - 2\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} + \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} \end{aligned} \quad (7)$$

where use is made of the properties of the transpose of a matrix, namely, $(\mathbf{X}\hat{\boldsymbol{\beta}})' = \hat{\boldsymbol{\beta}}'\mathbf{X}'$; and since $\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$ is a scalar (a real number), it is equal to its transpose $\mathbf{y}'\mathbf{X}\hat{\boldsymbol{\beta}}$.

OLS Estimation

$$\text{or } \frac{\partial}{\partial \beta} [(Y' - \hat{\beta}' X') (Y - X \hat{\beta})] = 0$$

$$\text{or } \frac{\partial}{\partial \hat{\beta}} \left[\underbrace{Y' Y}_{1 \times 1} - \underbrace{\hat{\beta}' X' Y}_{1 \times k \cdot k \times n \cdot n \times 1} - \underbrace{(Y' X \hat{\beta})}_{\substack{1 \times n \cdot n \times k \cdot k \times 1 \\ \downarrow \\ \hat{\beta}' X' Y}} + \hat{\beta}' X' X \hat{\beta} \right] = 0$$

$$\text{or } \frac{\partial}{\partial \hat{\beta}} \Rightarrow \left[\underbrace{0}_{k \times 1} - \underbrace{X' Y}_{k \times n \cdot n \times 1} - \underbrace{X' Y}_{k \times 1} + 2 \underbrace{X' X \hat{\beta}}_{k \times n \cdot n \times k \cdot k \times 1} \right] = 0$$

$$2 X' X \hat{\beta} = 2 X' Y$$

OLS Estimation

- Equation (7) is the matrix representation of (4). To minimize square sum of errors take derivative w.r.t β and equate to zero

$$(\mathbf{X}'\mathbf{X})\beta = \mathbf{X}'\mathbf{Y}$$

- Now using matrix algebra, if the inverse of $(\mathbf{X}'\mathbf{X})$ exists, say, $(\mathbf{X}'\mathbf{X})^{-1}$, then pre multiplying both sides of above equation by this inverse, we obtain

$$(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

But since $(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X}) = \mathbf{I}$, an identity matrix of order $k \times k$, we get

$$\mathbf{I}\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

or

$$\begin{array}{ccccc} \hat{\beta} & = & (\mathbf{X}'\mathbf{X})^{-1} & \mathbf{X}' & \mathbf{Y} \\ k \times 1 & & k \times k & (k \times n) & (n \times 1) \end{array}$$

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Example: 3 parametric case in matrix form or notation.

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + U$$

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X}_2 - \hat{\beta}_3 \bar{X}_3$$

In deviation form

$$y = \beta_2 x_2 + \beta_3 x_3 + u$$

where $y = Y - \bar{Y}$, $x_2 = X_2 - \bar{X}_2$, $x_3 = X_3 - \bar{X}_3$

$$u = U - \bar{U}$$

= 0

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$X'X =$$

$$\begin{bmatrix} x_{21} & x_{22} & \dots & x_{2n} \\ x_{31} & x_{32} & \dots & x_{3n} \end{bmatrix}$$

$$\begin{bmatrix} x_{21} & x_{31} \\ x_{22} & x_{32} \\ \vdots & \vdots \\ x_{2n} & x_{3n} \end{bmatrix}$$

$$\begin{bmatrix} \sum x_{2i}^2 & \sum x_{2i} x_{3i} \\ \sum x_{3i} x_{2i} & \sum x_{3i}^2 \end{bmatrix}$$

$$= \begin{bmatrix} \sum x_2^2 & \sum x_2 x_3 \\ \sum x_2 x_3 & \sum x_3^2 \end{bmatrix}$$

$$|X'X| = \underbrace{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}_{\text{Denote it by } \Delta}$$

Denote it by Δ

$$|X'X|^{-1} = \frac{\text{Adj} |X'X|}{|X'X|} = \frac{\begin{bmatrix} \sum x_3^2 & -\sum x_2 x_3 \\ -\sum x_2 x_3 & \sum x_2^2 \end{bmatrix}}{\Delta}$$

Now consider $X'Y$

$$X'Y = \begin{bmatrix} x_{21} & x_{22} & \dots & x_{2n} \\ x_{31} & x_{32} & \dots & x_{3n} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum x_{2y} \\ \sum x_{3y} \end{bmatrix}$$

Now $\hat{\beta}^* = (X'X)^{-1} X'Y$

Plug in values:

$$= \frac{1}{\Delta} \begin{bmatrix} \sum x_3^2 & -\sum x_2 \sum x_3 \\ -\sum x_2 x_3 & \sum x_2^2 \end{bmatrix} \begin{bmatrix} \sum x_{2y} \\ \sum x_{3y} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} \sum x_3^2 \sum x_{2y} - \sum x_2 x_3 \sum x_{3y} \\ -\sum x_2 x_3 \sum x_{2y} + \sum x_2^2 \sum x_{3y} \end{bmatrix}$$

So $\hat{\beta}_2 = \frac{\sum x_3^2 \sum x_{2y} - \sum x_2 x_3 \sum x_{3y}}{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}$

$$\hat{\beta}_3 = \frac{\sum x_2^2 \sum x_{3y} - \sum x_2 x_3 \sum x_{2y}}{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}$$

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}_2 - \hat{\beta}_3 \bar{x}_3$$

$$V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

where $V(\hat{\beta})$ is the var-cov matrix of $\hat{\beta}$

$$Var(\hat{\beta}) = \begin{bmatrix} Var(\hat{\beta}_1) & Cov(\hat{\beta}_1, \hat{\beta}_2) & \dots & Cov(\hat{\beta}_1, \hat{\beta}_k) \\ Cov(\hat{\beta}_2, \hat{\beta}_1) & Var(\hat{\beta}_2) & \dots & Cov(\hat{\beta}_2, \hat{\beta}_k) \\ \vdots & \vdots & & \vdots \\ Cov(\hat{\beta}_k, \hat{\beta}_1) & Cov(\hat{\beta}_k, \hat{\beta}_2) & \dots & Var(\hat{\beta}_k) \end{bmatrix}$$

Var - Cov matrix.

$$V(\beta) = \sigma_u^2 (X'X)^{-1} = \sigma^2 \begin{bmatrix} \sum x_3^2 & -\sum x_2 x_3 \\ -\sum x_2 x_3 & \sum x_2^2 \end{bmatrix} / \Delta$$

$$\begin{bmatrix} \text{Var}(\hat{\beta}_2) & \text{Cov}(\hat{\beta}_2, \hat{\beta}_3) \\ \text{Cov}(\hat{\beta}_3, \hat{\beta}_2) & \text{Var}(\hat{\beta}_3) \end{bmatrix} = \begin{bmatrix} \sigma^2 \sum x_3^2 / \Delta & -\sigma^2 \sum x_2 x_3 / \Delta \\ -\sigma^2 \sum x_2 x_3 / \Delta & \sigma^2 \sum x_2^2 / \Delta \end{bmatrix}$$

$$\text{Var}(\hat{\beta}_2) = \sigma_u^2 \frac{\sum x_3^2}{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}$$

$$\text{Var}(\hat{\beta}_3) = \sigma_u^2 \frac{\sum x_2^2}{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}$$

$$\text{Cov}(\hat{\beta}_2, \hat{\beta}_3) = -\sigma_u^2 \frac{\sum x_2 x_3}{\sum x_2^2 \sum x_3^2 - (\sum x_2 x_3)^2}$$

R^2 and the Adjusted R^2

An important property of R^2 is that it is a nondecreasing function of the number of explanatory variables or regressors present in the model, unless the added variable is perfectly collinear with the other regressors; as the number of regressors increases, R^2 almost invariably increases and never decreases. Stated differently, an additional X variable will not decrease R^2 . To see this, recall the definition of the coefficient of determination:

$$\begin{aligned} R^2 &= \frac{\text{ESS}}{\text{TSS}} \\ &= 1 - \frac{\text{RSS}}{\text{TSS}} \\ &= 1 - \frac{\sum \hat{u}_i^2}{\sum y_i^2} \end{aligned} \tag{7.8.1}$$

Now $\sum y_i^2$ is independent of the number of X variables in the model because it is simply $\sum (Y_i - \bar{Y})^2$. The RSS, $\sum \hat{u}_i^2$, however, depends on the number of regressors present in the model. Intuitively, it is clear that as the number of X variables increases, $\sum \hat{u}_i^2$ is likely to decrease (at least it will not increase); hence R^2 as defined in Eq. (7.8.1) will increase. In view of this, in comparing two regression models with the *same dependent variable* but differing number of X variables, one should be very wary of choosing the model with the highest R^2 .

To compare two R^2 terms, one must take into account the number of X variables present in the model. This can be done readily if we consider an alternative coefficient of determination, which is as follows:

$$\bar{R}^2 = 1 - \frac{\sum \hat{u}_i^2 / (n - k)}{\sum y_i^2 / (n - 1)} \quad (7.8.2)$$

where k = the number of parameters in the model *including the intercept term*. (In the three-variable regression, $k = 3$. Why?) The R^2 thus defined is known as the **adjusted R^2** , denoted by $\bar{R}^2$. The term *adjusted* means adjusted for the df associated with the sums of squares entering into Eq. (7.8.1): $\sum \hat{u}_i^2$ has $n - k$ df in a model involving k parameters, which include the intercept term, and $\sum y_i^2$ has $n - 1$ df. (Why?) For the three-variable case, we know that $\sum \hat{u}_i^2$ has $n - 3$ df.

Equation (7.8.2) can also be written as

$$\bar{R}^2 = 1 - \frac{\hat{\sigma}^2}{S_Y^2} \quad (7.8.3)$$

where $\hat{\sigma}^2$ is the residual variance, an unbiased estimator of true σ^2 , and S_Y^2 is the sample variance of Y .

It is easy to see that $\bar{R}^2$ and R^2 are related because, substituting Eq. (7.8.1) into Eq. (7.8.2), we obtain

$$\bar{R}^2 = 1 - (1 - R^2) \frac{n - 1}{n - k} \quad (7.8.4)$$

It is immediately apparent from Eq. (7.8.4) that (1) for $k > 1$, $\bar{R}^2 < R^2$ which implies that as the number of X variables increases, the adjusted R^2 increases less than the unadjusted R^2 ; and (2) $\bar{R}^2$ can be negative, although R^2 is necessarily nonnegative.¹⁰ In case $\bar{R}^2$ turns out to be negative in an application, its value is taken as zero.

Which R^2 should one use in practice? As Theil notes:

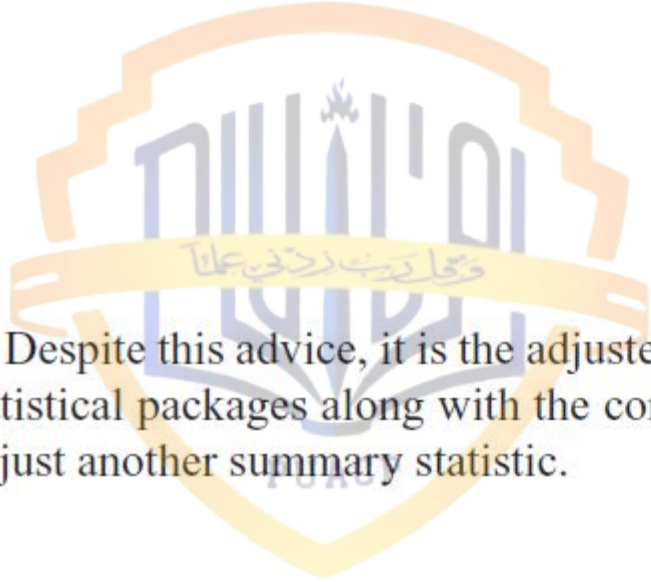
... it is good practice to use $\bar{R}^2$ rather than R^2 because R^2 tends to give an overly optimistic picture of the fit of the regression, particularly when the number of explanatory variables is not very small compared with the number of observations.¹¹

But Theil's view is not uniformly shared, for he has offered no general theoretical justification for the "superiority" of $\bar{R}^2$. For example, Goldberger argues that the following R^2 , call it **modified** R^2 , will do just as well:¹²

$$\text{Modified } R^2 = (1 - k/n)R^2 \quad (7.8.5)$$

His advice is to report R^2 , n , and k and let the reader decide how to adjust R^2 by allowing for n and k .

¹⁰Note, however, that if $R^2 = 1$, $\bar{R}^2 = R^2 = 1$. When $R^2 = 0$, $\bar{R}^2 = (1 - k)/(n - k)$, in which case $\bar{R}^2$ can be negative if $k > 1$.



Despite this advice, it is the adjusted R^2 , as given in Eq. (7.8.4), that is reported by most statistical packages along with the conventional R^2 . The reader is well advised to treat $\bar{R}^2$ as just another summary statistic.

Besides R^2 and adjusted R^2 as goodness of fit measures, other criteria are often used to judge the adequacy of a regression model. Two of these are **Akaike's Information criterion** and **Amemiya's Prediction criteria**, which are used to select between competing models. We will discuss these criteria when we consider the problem of model selection in greater detail in a later chapter (see Chapter 13).

Analysis of Variance

Source of Variation	SS*	df	MSS†
Due to regression (ESS)	$\sum \hat{y}_i^2 = \hat{\beta}_2^2 \sum x_i^2$	1	$\hat{\beta}_2^2 \sum x_i^2$
Due to residuals (RSS)	$\sum \hat{u}_i^2$	$n - 2$	$\frac{\sum \hat{u}_i^2}{n - 2} = \hat{\sigma}^2$
TSS	$\sum y_i^2$	$n - 1$	

*SS means sum of squares.

†Mean sum of squares, which is obtained by dividing SS by their df.

$$\begin{aligned}
 F &= \frac{\text{MSS of ESS}}{\text{MSS of RSS}} \\
 &= \frac{\hat{\beta}_2^2 \sum x_i^2}{\sum \hat{u}_i^2 / (n - 2)} \\
 &= \frac{\hat{\beta}_2^2 \sum x_i^2}{\hat{\sigma}^2}
 \end{aligned}$$

The Analysis of Variance Approach to Testing the Overall Significance of an Observed Multiple Regression: The F Test

$$\sum y_i^2 = \hat{\beta}_2 \sum y_i x_{2i} + \hat{\beta}_3 \sum y_i x_{3i} + \sum \hat{u}_i^2$$

TSS = ESS + RSS

Source of Variation	SS	df	MSS
Due to regression (ESS)	$\hat{\beta}_2 \sum y_i x_{2i} + \hat{\beta}_3 \sum y_i x_{3i}$	2	$\frac{\hat{\beta}_2 \sum y_i x_{2i} + \hat{\beta}_3 \sum y_i x_{3i}}{2}$
Due to residual (RSS)	$\sum \hat{u}_i^2$	$n - 3$	$\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n - 3}$
Total	$\sum y_i^2$	$n - 1$	

$$F = \frac{(\hat{\beta}_2 \sum y_i x_{2i} + \hat{\beta}_3 \sum y_i x_{3i}) / 2}{\sum \hat{u}_i^2 / (n - 3)} = \frac{\text{ESS/df}}{\text{RSS/df}}$$

is distributed as the F distribution with 2 and $n - 3$ df.

Testing the Overall Significance of a Multiple Regression: The F Test

Decision Rule

Given the k -variable regression model:

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \cdots + \beta_k X_{ki} + u_i$$

To test the hypothesis

$$H_0: \beta_2 = \beta_3 = \cdots = \beta_k = 0$$

(i.e., all slope coefficients are simultaneously zero) versus

$$H_1: \text{Not all slope coefficients are simultaneously zero}$$

compute

$$F = \frac{\text{ESS}/\text{df}}{\text{RSS}/\text{df}} = \frac{\text{ESS}/(k-1)}{\text{RSS}/(n-k)} \quad (8.4.7)$$

If $F > F_\alpha(k-1, n-k)$, reject H_0 ; otherwise you do not reject it, where $F_\alpha(k-1, n-k)$ is the *critical F* value at the α level of significance and $(k-1)$ numerator df and $(n-k)$ denominator df. Alternatively, if the p value of F obtained from Eq. (8.4.7) is sufficiently low, one can reject H_0 .

An Important Relationship between R^2 and F

There is an intimate relationship between the coefficient of determination R^2 and the F test used in the analysis of variance. Assuming the normal distribution for the disturbances u_i and the null hypothesis that $\beta_2 = \beta_3 = 0$, we have seen that

$$F = \frac{\text{ESS}/2}{\text{RSS}/(n-3)} \quad (8.4.8)$$

is distributed as the F distribution with 2 and $n - 3$ df.

More generally, in the k -variable case (including intercept), if we assume that the disturbances are normally distributed and that the null hypothesis is

$$H_0: \beta_2 = \beta_3 = \cdots = \beta_k = 0 \quad (8.4.9)$$

then it follows that

$$F = \frac{\text{ESS}/(k-1)}{\text{RSS}/(n-k)} \quad (8.4.7) = (8.4.10)$$

follows the F distribution with $k - 1$ and $n - k$ df. (Note: The total number of parameters to be estimated is k , of which 1 is the intercept term.)

An Important Relationship between R^2 and F

Let us manipulate Eq. (8.4.10) as follows:

$$\begin{aligned} F &= \frac{n - k}{k - 1} \frac{\text{ESS}}{\text{RSS}} \\ &= \frac{n - k}{k - 1} \frac{\text{ESS}}{\text{TSS} - \text{ESS}} \\ &= \frac{n - k}{k - 1} \frac{\text{ESS}/\text{TSS}}{1 - (\text{ESS}/\text{TSS})} \\ &= \frac{n - k}{k - 1} \frac{R^2}{1 - R^2} \\ &= \frac{R^2/(k - 1)}{(1 - R^2)/(n - k)} \end{aligned}$$

(8.4.11)

Testing the Overall Significance of a Multiple Regression: The F Test

Decision Rule

Testing the overall significance of a regression in terms of R^2 : Alternative but equivalent test to Eq. (8.4.7).

Given the k -variable regression model:

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \cdots + \beta_k X_{ki} + u_i$$

To test the hypothesis

$$H_0: \beta_2 = \beta_3 = \cdots = \beta_k = 0$$

versus

H_1 : Not all slope coefficients are simultaneously zero

compute

$$F = \frac{R^2/(k-1)}{(1-R^2)/(n-k)} \quad (8.4.13)$$

If $F > F_{\alpha(k-1, n-k)}$, reject H_0 ; otherwise you may accept H_0 where $F_{\alpha(k-1, n-k)}$ is the critical F value at the α level of significance and $(k-1)$ numerator df and $(n-k)$ denominator df. Alternatively, if the p value of F obtained from Eq. (8.4.13) is sufficiently low, reject H_0 .

The “Incremental” or “Marginal” Contribution of an Explanatory Variable

$$F = \frac{(\text{ESS}_{\text{new}} - \text{ESS}_{\text{old}}) / \text{number of new regressors}}{\text{RSS}_{\text{new}} / \text{df} (= n - \text{number of parameters in the new model})}$$

$$\begin{aligned} F &= \frac{(R_{\text{new}}^2 - R_{\text{old}}^2) / \text{df}}{(1 - R_{\text{new}}^2) / \text{df}} \\ &= \frac{(R_{\text{new}}^2 - R_{\text{old}}^2) / \text{number of new regressors}}{(1 - R_{\text{new}}^2) / \text{df} (= n - \text{number of parameters in the new model})} \end{aligned}$$

Testing the Equality of Two Regression Coefficients

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + u_i \quad (8.5.1)$$

$$H_0: \beta_3 = \beta_4 \quad \text{or} \quad (\beta_3 - \beta_4) = 0 \quad (8.5.2)$$

$$H_1: \beta_3 \neq \beta_4 \quad \text{or} \quad (\beta_3 - \beta_4) \neq 0$$

$$t = \frac{(\hat{\beta}_3 - \hat{\beta}_4) - (\beta_3 - \beta_4)}{\text{se}(\hat{\beta}_3 - \hat{\beta}_4)} \quad (8.5.3)$$

$$\text{se}(\hat{\beta}_3 - \hat{\beta}_4) = \sqrt{\text{var}(\hat{\beta}_3) + \text{var}(\hat{\beta}_4) - 2 \text{cov}(\hat{\beta}_3, \hat{\beta}_4)} \quad (8.5.4)$$

$$t = \frac{\hat{\beta}_3 - \hat{\beta}_4}{\sqrt{\text{var}(\hat{\beta}_3) + \text{var}(\hat{\beta}_4) - 2 \text{cov}(\hat{\beta}_3, \hat{\beta}_4)}} \quad (8.5.5)$$

The *F*-Test Approach: Restricted Least Squares

The preceding *t* test is a kind of postmortem examination because we try to find out whether the linear restriction is satisfied after estimating the “unrestricted” regression. A direct approach would be to incorporate the restriction (8.6.3) into the estimating procedure at the outset. In the present example, this procedure can be done easily. From (8.6.3) we see that

$$\beta_2 = 1 - \beta_3 \quad (8.6.5)$$

or

$$\beta_3 = 1 - \beta_2 \quad (8.6.6)$$

Therefore, using either of these equalities, we can eliminate one of the β coefficients in Eq. (8.6.2) and estimate the resulting equation. Thus, if we use Eq. (8.6.5), we can write the Cobb–Douglas production function as

$$\begin{aligned} \ln Y_i &= \beta_0 + (1 - \beta_3) \ln X_{2i} + \beta_3 \ln X_{3i} + u_i \\ &= \beta_0 + \ln X_{2i} + \beta_3 (\ln X_{3i} - \ln X_{2i}) + u_i \end{aligned}$$

or

$$(\ln Y_i - \ln X_{2i}) = \beta_0 + \beta_3 (\ln X_{3i} - \ln X_{2i}) + u_i \quad (8.6.7)$$

or

$$\ln (Y_i/X_{2i}) = \beta_0 + \beta_3 \ln (X_{3i}/X_{2i}) + u_i \quad (8.6.8)$$

where (Y_i/X_{2i}) = output/labor ratio and (X_{3i}/X_{2i}) = capital labor ratio, quantities of great economic importance.

The F -Test Approach: Restricted Least Squares

- How do we compare the unrestricted and restricted least-squares regressions? This question can be answered by applying the F test as follows. Let

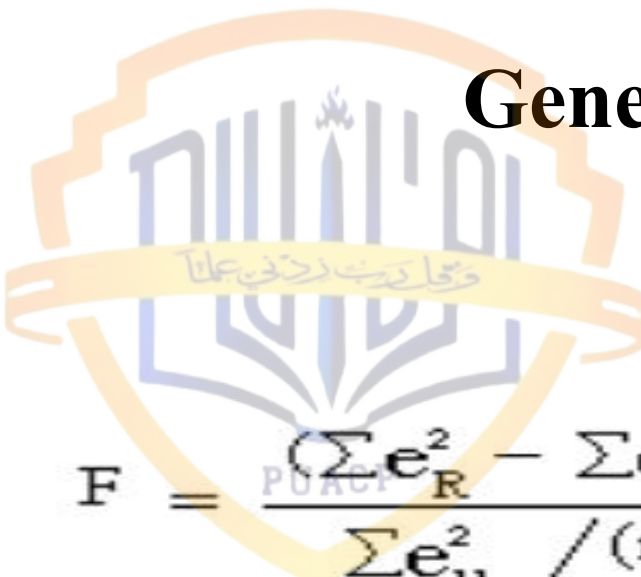
$$\begin{aligned} F &= \frac{(RSS_R - RSS_{UR})/m}{RSS_{UR}/(n - k)} \\ &= \frac{(\sum \hat{u}_R^2 - \sum \hat{u}_{UR}^2)/m}{\sum \hat{u}_{UR}^2/(n - k)} \end{aligned} \quad (8.6.9)$$

follows the F distribution with m , $(n - k)$ df. (Note: UR and R stand for unrestricted and restricted, respectively.)

The F test above can also be expressed in terms of R^2 as follows:

$$F = \frac{(R_{UR}^2 - R_R^2)/m}{(1 - R_{UR}^2)/(n - k)} \quad (8.6.10)$$

General F Testing


$$F = \frac{(\sum e_R^2 - \sum e_u^2) / R}{\sum e_u^2 / (n - k)}$$

Where

$\sum e_R^2$ = Restricted residual sum of squares.

$\sum e_u^2$ = Unrestricted residual sum of squares.

R = Number of restrictions.

n = Number of observations.

k = Number of parameters in unrestricted model.

The Steps are as follows:

- 1). Estimate the unrestricted equation to compute $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4$.

Then compute

$$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3 + \hat{\beta}_4 X_4$$

Then

$$e_i = Y_i - \hat{Y}_i.$$

Finally compute $\sum e_u^2$

Suppose $\sum e_u^2 = 50$.

- 2). Impose the restriction given into this will yield, according to our example;

$$Y = \beta_1 + (1) X_2 + (0) X_3 + \beta_4 X_4 + U$$

$$Y - X_2 = \beta_1 + \beta_4 X_4 + U$$

Estimate β_1 and β_4 ; and compute;

$$\hat{\beta}_1 = \dots\dots\dots$$

$$\hat{\beta}_4 = \dots\dots\dots$$

Now complete

$$\hat{Y} = \hat{\beta}_1 + X_2 + \hat{\beta}_4 X_4$$

$$e_i = Y_i - \hat{Y}_i.$$

Finally compute $\sum e_R^2$

Suppose we have $\sum e_R^2 = 60$.

Multiple Regression Model:

→ **Testing of more than one restriction jointly:**

Suppose we have to estimate or test.

$$H_0: \beta_2 = 1, \beta_3 = 0$$

$$H_1: \beta_2 \neq 1 \text{ and/ or } \beta_3 \neq 0,$$

And one model is

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + U$$

→ [We have to see the number of restrictions and not to focus on how many restrictions on parameters].

$$F = \frac{(\sum e_R^2 - \sum e_u^2) / R}{\sum e_u^2 / (n - k)}$$

Where

$\sum e_R^2$ = Restricted residual sum of squares.

$\sum e_u^2$ = Unrestricted residual sum of squares.

R = Number of restrictions.

n = Number of observations.

k = Number of parameters in unrestricted model.

3). Compute the F-statistics.

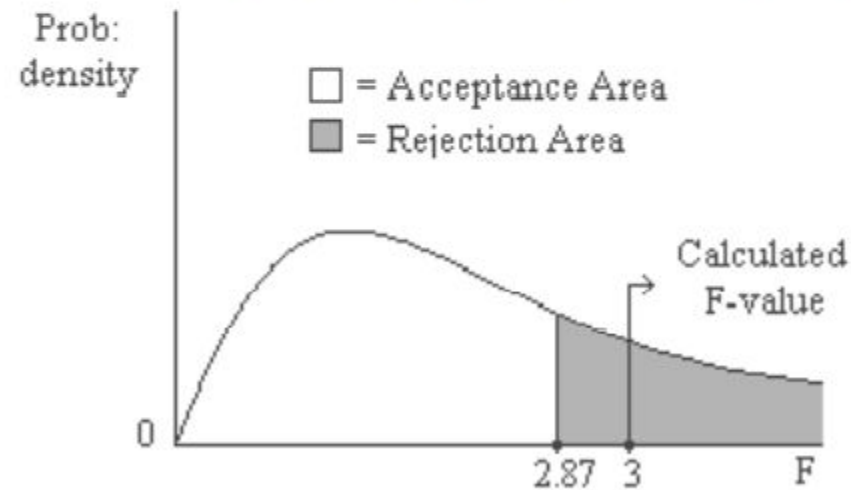
Note these values; $\sum e_u^2 = 50$, $\sum e_R^2 = 60$, $R = 2$, $n = 34$ and $k = 4$.

Now plug in values;

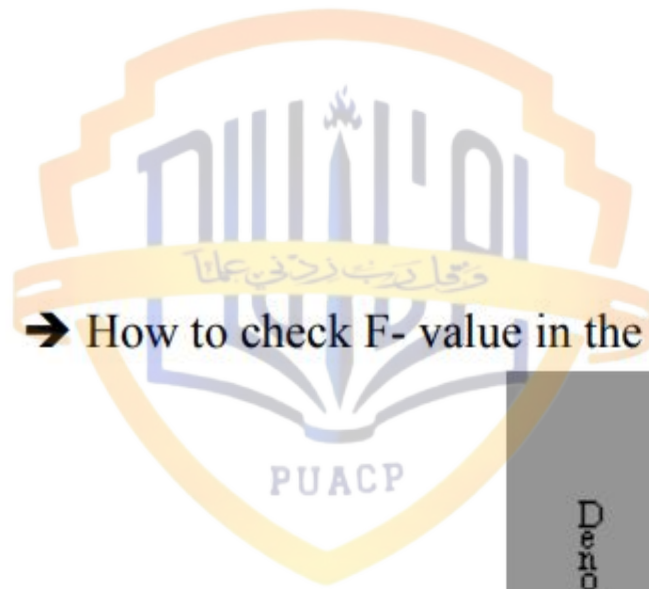
$$F = \frac{(60 - 50) / 2}{50 / (34 - 4)} = \frac{10 / 2}{50 / 30} = \frac{5 * 30}{50} = 3.$$

4). Conclusion.

We conclude by comparing calculated F-value with the critical F-value. In our case the critical F-value at $R = 2$ and $n - k = 30$ and $df = 2.87$ is supposed.



→ In our example the calculated F-value > critical F-value, so we reject H_0 .



→ How to check F- value in the Table:

	Numerator (R = 2)			
	1	2	3	4
Denominator (n - k)				
2				
3				
4				
...				
30				

→ Consider now General model:

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \dots \beta_K X_K + U$$

Special Case 1: Just one restriction.

Examples: $H_0: \beta_1 = 0.$

$H_0: \beta_1 = 1.$

$H_0: \beta_2 + \beta_3 = -1.$

$H_0: \beta_2 + \beta_3 + \beta_4 + \beta_5 = 0.$

$H_0: \beta_2 + \beta_3 + \beta_4 + \dots + \beta_K = 0.$

→ In this case we can show that;

$$F = t^2$$

$$\text{Critical } F = (\text{critical } t)^2$$

$$|t| > |t_{\text{critical}}| \Leftrightarrow |F| > |F_{\text{critical}}|$$

$$|t| = |t_{\text{critical}}| \Leftrightarrow |F| = |F_{\text{critical}}|$$

$$|t| < |t_{\text{critical}}| \Leftrightarrow |F| < |F_{\text{critical}}|$$

→ So “t” and “F” test will lead to identical conclusions.

TESTING THE FUNCTIONAL FORM OF REGRESSION: CHOOSING BETWEEN LINEAR AND LOG-LINEAR REGRESSION MODELS

The MWD test involves the following steps:²¹

Step I: Estimate the linear model and obtain the estimated Y values. Call them Yf (i.e., $\hat{Y}$).

Step II: Estimate the log-linear model and obtain the estimated $\ln Y$ values; call them $\ln f$ (i.e., $\widehat{\ln Y}$).

Step III: Obtain $Z_1 = (\ln Yf - \ln f)$.

Step IV: Regress Y on X 's and Z_1 obtained in Step III. Reject H_0 if the coefficient of Z_1 is statistically significant by the usual t test.

Step V: Obtain $Z_2 = (\text{antilog of } \ln f - Yf)$.

Step VI: Regress log of Y on the logs of X 's and Z_2 . Reject H_1 if the coefficient of Z_2 is statistically significant by the usual t test.

Special Case 2: Each parameter except intercept is set equal to zero.

$$H_0: \beta_2 = 0, \beta_3 = 0, \beta_4 = 0 \dots\dots\dots \beta_K = 0.$$

$$H_1: \text{At least one } \beta_j \neq 0, [j = 2, 3, 4 \dots\dots\dots k]$$

→ In this case restricted model becomes;

$$Y = \beta_1 + U$$

Or

$$Y = b_1 + V$$

$$\hat{b}_1 = \bar{Y}$$

$$\hat{Y} = \hat{b}_1 = \bar{Y}$$

$$e_R = Y - \hat{Y} = Y - \bar{Y} = y$$

Now

$$F = \frac{(\sum e_R^2 - \sum e_u^2) / R}{\sum e_u^2 / (n - k)}$$

$$\begin{aligned}
 &= \frac{(\sum \hat{y}^2 - \sum e_u^2) / (k - 1)}{\sum e_u^2 / (n - k)} \\
 &= \frac{(\sum \hat{y}^2 - \sum e^2) / (k - 1)}{\sum e^2 / (n - k)}
 \end{aligned}$$

→ Where all values come from unrestricted model, so we can ignore the subscript $\bar{u}$.

We can write model as;

$$F = \frac{\sum \hat{y}^2 / (k - 1)}{\sum e^2 / (n - k)}$$

→ It is test of over all performance of the model.

Note 1: In General Case.

$$F = \frac{(\sum e_R^2 - \sum e_u^2) / R}{\sum e_u^2 / (n - k)}$$

Or

$$= \frac{[(\sum y^2 - \sum \hat{y}_R^2) - (\sum y^2 - \sum \hat{y}_u^2)] / R}{(\sum y^2 - \sum \hat{y}_u^2) / (n - k)}$$

Divide all terms by $\sum y^2$.

$$= \frac{\left[1 - \frac{\sum \hat{y}_R^2}{\sum y^2}\right] - \left[1 - \frac{\sum \hat{y}_u^2}{\sum y^2}\right] / R}{\left[1 - \frac{\sum \hat{y}_u^2}{\sum y^2}\right] / (n - k)}$$
$$= \frac{[R_u^2 - R_R^2] / R}{[1 - R_u^2] / (n - k)}$$

→ F-statistics indicates that increase in the restriction decrease R^2 by more error (In against or alternative).

→ F-statistics indicates the increase in the R^2 due to removal of restrictions.

Note 2: In Special Case 2.

$$F = \frac{\sum \hat{y}^2 / (k - 1)}{\sum e^2 / (n - k)}$$

Divide all terms by $\sum y^2$

$$\begin{aligned} & \frac{\sum \hat{y}^2 / (k - 1)}{\sum y^2} \\ &= \frac{\sum e^2 / (n - k)}{\sum y^2} \end{aligned}$$

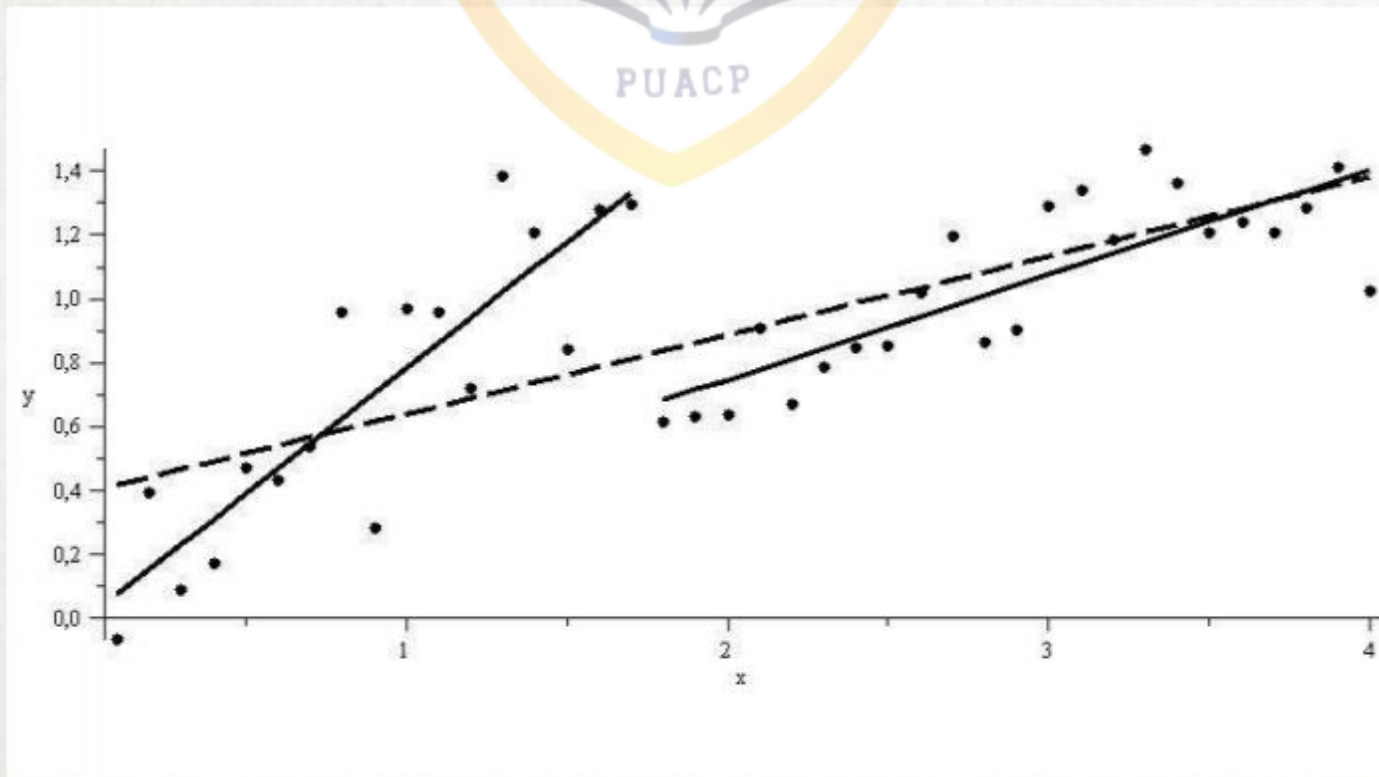
$$= \frac{R^2 / (k - 1)}{[1 - R^2] / (n - k)}$$

When $F = 0 \Rightarrow R^2 = 0$ test just like.

→ F-statistics indicates the significance difference of R^2 from zero.

TESTING FOR STRUCTURAL OR PARAMETER STABILITY OF REGRESSION MODELS: THE CHOW TEST

- When we use a regression model involving time series data, it may happen that there is a **structural change** in the relationship between the regressand Y and the regressors.



CHOW TEST

Suppose that we model our data as

$$y_t = a + bx_{1t} + cx_{2t} + \varepsilon_t.$$

If we split our data into two groups, then we have

$$y_t = a_1 + b_1x_{1t} + c_1x_{2t} + \varepsilon_{1t}$$

and

$$y_t = a_2 + b_2x_{1t} + c_2x_{2t} + \varepsilon_t.$$

The **null hypothesis** of the Chow test asserts that $a_1 = a_2$, $b_1 = b_2$, and $c_1 = c_2$, and there is the assumption that the **model errors** ε are **independent and identically distributed** from a **normal distribution** with unknown **variance**.

$$F = \frac{(\text{RSS}_R - \text{RSS}_{UR})/k}{(\text{RSS}_{UR})/(n_1 + n_2 - 2k)} \sim F_{[k, (n_1 + n_2 - 2k)]} \quad (8.7.4)$$

TESTING THE FUNCTIONAL FORM OF REGRESSION: CHOOSING BETWEEN LINEAR AND LOG-LINEAR REGRESSION MODELS

- The choice between a linear regression model (the regressand is a linear function of the regressors) or a log-linear regression model (the log of the regressand is a function of the logs of the regressors) is a perennial question in empirical analysis. We can use a test proposed by MacKinnon, White, and Davidson, which for brevity we call the **MWD test**, to choose between the two models.

H_0 : *Linear Model*: Y is a linear function of regressors, the X 's.

H_1 : *Log-Linear Model*: $\ln Y$ is a linear function of logs of regressors, the logs of X 's.

TESTING THE FUNCTIONAL FORM OF REGRESSION: CHOOSING BETWEEN LINEAR AND LOG-LINEAR REGRESSION MODELS

- If the linear model is in fact the correct model, the constructed variable $Z1$ should not be statistically significant in Step IV, for in that case the estimated Y values from the linear model and those estimated from the log-linear model (after taking their antilog values for comparative purposes) should not be different. The same comment applies to the alternative hypothesis $H1$.

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